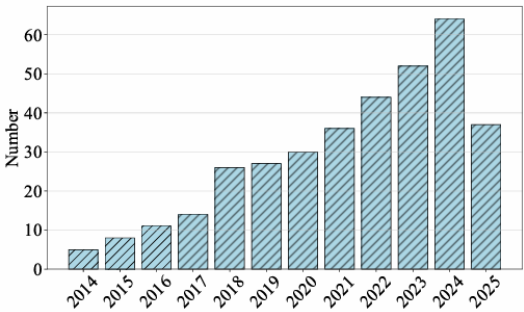
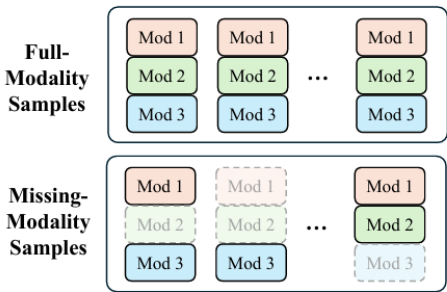


1. Introduction

Multimodal learning integrates diverse data streams—such as visual, textual, auditory, and biological sensory signals—to emulate human multi-sensory perception. By modeling correlations and complementarities across different channels, multimodal architectures reveal deep patterns unavailable to unimodal baselines. However, deploying these models in real-world environments introduces a critical challenge: Missing Modality (or Incomplete Modality) during training, inference, or both phases.



(a) Trends in Missing Modality Publications



(b) Full-Modality vs. Missing-Modality Samples

Primary Factors Inducing Modality Unavailability:

Sensor Failures & Hardware Degradation: Physical sensors frequently malfunction or fail entirely due to environmental stress. For example, severe temperature shifts caused the loss of the inclinometer on the Mars Ingenuity helicopter, forcing reliance on fallback telemetry.

Privacy Barriers & High Costs: In clinical settings, acquiring certain high-fidelity modalities (e.g., MRI or PET scans) may be restricted by financial limits, emergency time constraints, or patient data-sharing regulations

Environmental Noise & Transmission Losses: Real-time streaming applications suffer from random packet drops, acoustic occlusion, or lens obstruction, degrading continuous information flow.

Structural Heterogeneity: Discrepancies between historical data sources or hardware updates often result in structural missing patterns where specific modalities are systematically omitted across portions of the dataset

2 Methodology Taxonomy

The academic landscape of MLMM incorporates solutions divided into two main dimensions: Data Processing (operating at the raw input or feature level) and Strategy Design (operating at the model architecture or systemic optimization level).

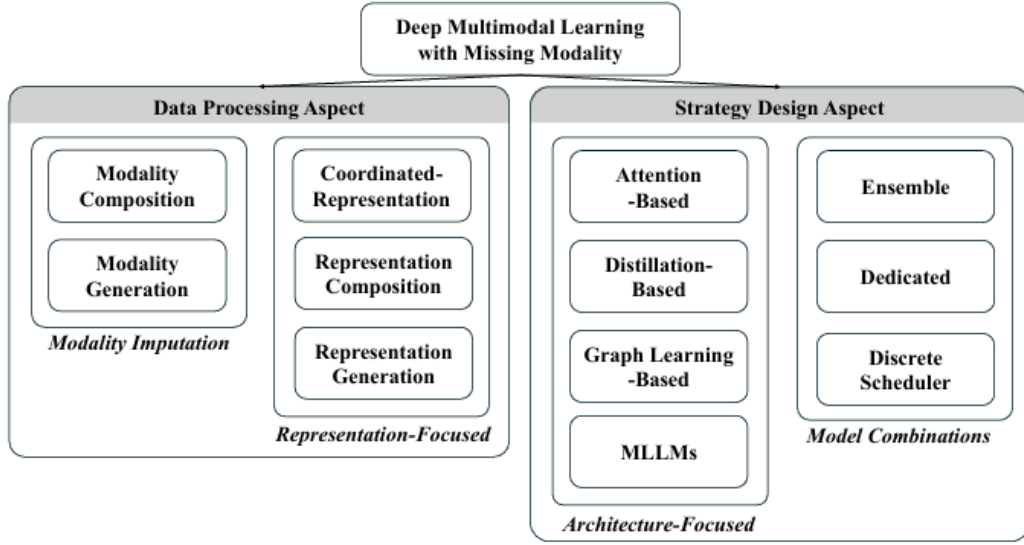


Figure 2: Our taxonomy of deep multimodal learning with missing modality methods. We categorize existing methods into two aspects: data processing and strategy design. Data Processing: we differentiate between modality imputation (handling at the modality data level) and representation-focused models (dealing with at the data representation level). Strategy Design: we distinguish between architecture-focused models (model architecture adjustments) and model combinations (combining multiple models externally). “MLLMs”: multimodal large language models.

Deep Dive into Data Processing

3.1 Modality Imputation (Raw-Level Completion)

Imputation methodologies intervene directly at the input level, transforming incomplete multimodal sets into complete matrices before feeding them into standard downstream models.

3.1.1 Modality Composition Methods

Zero or Random Value Imputation: Absent feature entries are substituted with static zeros or random noise distributions. This acts as a baseline, forcing the network to discover alternative pathways and reducing strict reliance on a single dominant modality. However, it fails in pixel-level tasks such as dense medical segmentation.

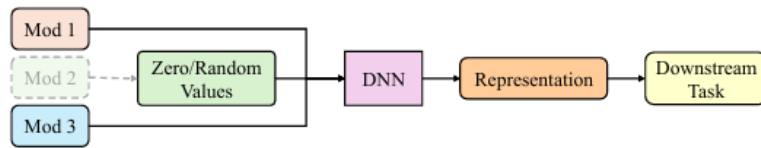


Figure 3: Zero/Random values composition methods. If we assume modality 2 is missing, then this modality will be replaced with zero/random values. “DNN” in all figures of this survey means different kinds of deep neural networks.

Retrieval-Based & Frame-Repeat Mechanisms: In sequential streaming contexts (such as audio-visual speech processing), missing segments are filled by repeating the preceding valid frame. Alternatively, K-Nearest Neighbors (K-NN) structures retrieve and copy corresponding data from a close semantic instance within the dataset. While functional, it induces massive memory overhead and struggles with fine-grained temporal alignment.

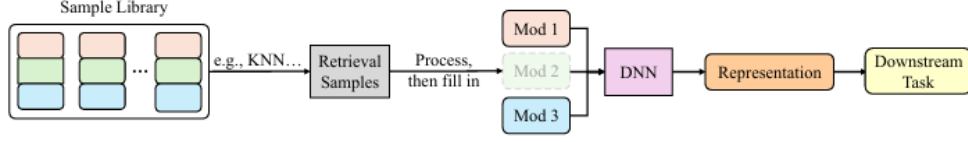
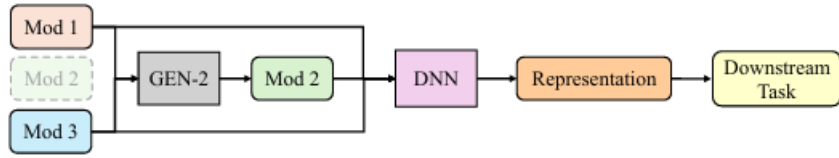
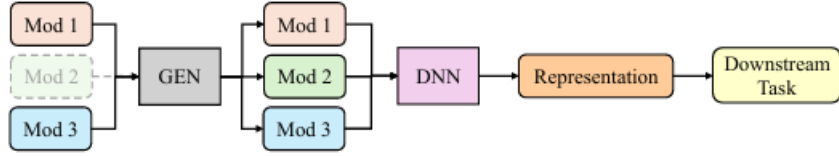


Figure 4: Retrieval-based modality composition methods search for one or more samples by randomly selecting or using simple retrieval algorithms like KNN, or its variants, from same-category samples that have the required missing modalities, and then compose them with the input missing-modality sample to form a “full”-modality sample.



(a) Individual Modality Generation Methods



(b) Unified Modality Generation Methods

Figure 5: Description of two typical modality generation methods. We set modality 2 as the missing modality for examples and use other available modalities to generate modality 2. “GEN” in both figures represents modality generation networks. (a) We set up a modality-2 generator (GEN-2) from other modalities. (b) All modalities are input and generated together by using a single GEN.

3.1.2 Modality Generative Methods

Individual Generative Models: Employs dedicated sub-networks (e.g., GANs, Autoencoders, or Diffusion Models) assigned to synthesize specific missing streams conditioned on the remaining observed modalities.

Unified Generative Frameworks: Implements a centralized generative architecture (e.g., Cascaded Autoencoders) capable of accepting arbitrary available inputs to synthesize all missing channels concurrently in a single pass, avoiding parameter inflation.

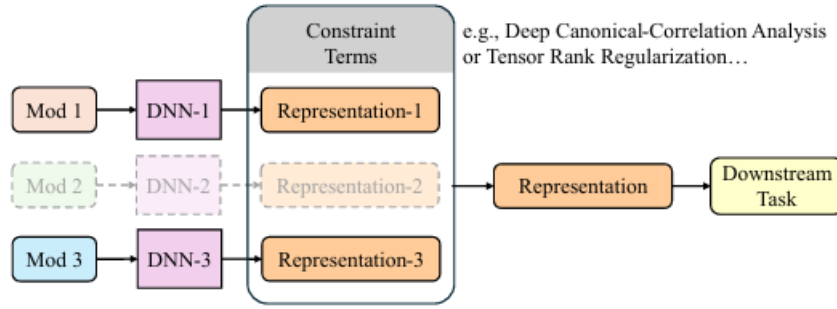


Figure 6: Illustration of coordinated-representation methods, which impose constraints to make the learned representations semantically consistent. Assuming that modality 2 is missing, these methods use constraints terms between the features of different modalities.

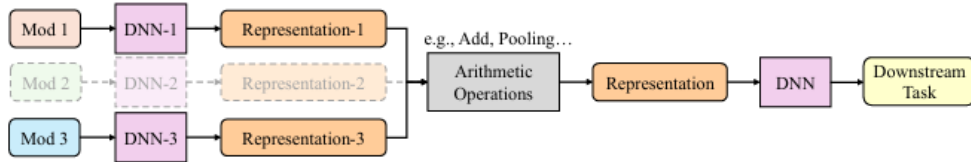


Figure 7: The general idea of the arithmetic operation-based representation composition methods. The representation of any number of modalities can be combined to a fixed dimension that is the same as the dimension required by the subsequent DNN layer.

3.2 Representation-Focused Frameworks (Latent-Level Completion)

Instead of rebuilding high-dimensional, noisy raw inputs, these approaches perform alignment and completion inside the hidden embedding space, saving computational resources.

Coordinated Representations: Enforces mathematical regularization constraints (e.g., maximizing mutual information or minimizing contrastive loss) during joint training to project all modalities into a tightly unified, shared latent space where missingness does not distort semantic retrieval.

Representation Composition: Aggregates latent vectors from available channels using static algebraic operations (such as max-pooling, mean-pooling, or concatenation) to build a robust multimodal token without adding trainable parameters.

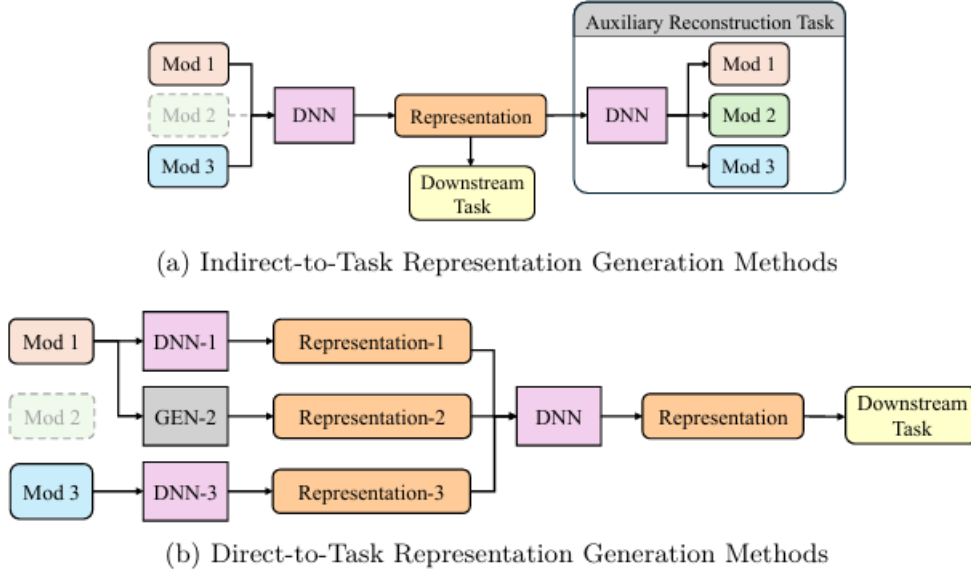


Figure 8: Description of two typical representation generation methods. We assume modality 2 is missing here. (a) Indirect-to-task representation generation methods are supervised by the loss of two tasks, namely: auxiliary reconstruction and downstream tasks. Note that these two tasks sometimes can be trained separately, first training the reconstruction task, discarding the generator, and then training the downstream task, such as the training progress of ActionMAE (Woo et al., 2023). While (b) direct-to-task representation generation methods directly use a modality generation network (“GEN-2” in figure) to generate representations of the missing modality 2 during training and inference without reconstructing the modality 2.

Representation Generation:

Indirect-to-Task: Latent reconstruction is treated as an auxiliary training target. The model learns to complete missing feature vectors, but the reconstruction layer is dropped during final inference.

Direct-to-Task: A specialized module learns a direct mapping function from the embeddings of available inputs to the missing modality embeddings, optimized purely for downstream performance rather than visual fidelity.

Deep Dive into Strategy Design

4.1 Architecture-Focused Paradigms

These methods restructure the internal operations of neural networks to natively tolerate dynamic input dropouts.

4.1.1 Attention-Based Architectures

Leveraging Transformer topologies, these systems employ dynamic self-attention or cross-attention layers. When a modality goes missing, the system applies an attention mask (setting missing token values to negative infinity before the softmax layer). This completely prevents the model from assigning weight to the missing signals without breaking the structural pipeline.

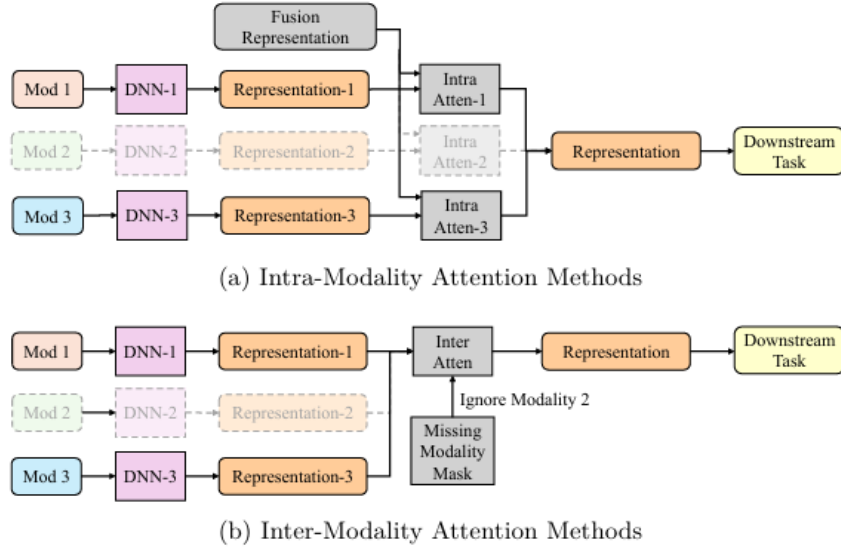


Figure 9: Two typical attention fusion methods, assuming modality 2 as the missing modality. (a) The intra-modality attention (“Intra-Atten” in figure) of missing modality 2 can be skipped. Since each Intra Atten of different modalities shares the fusion representation (like learnable query in (Ge et al., 2023) or bottleneck tokens in (Nagrani et al., 2021)), after calculating them one by one, the output representation can be regarded as the fusion of available modalities. (b) “Missing Modality Mask” is the custom mask that is generated according to the missing modality 2 and can help inter-modality attention (“Inter-Atten” in figure) ignore the missing modality 2. By setting the tokens of modality 2 to zero or negative infinity through masking, we can force the attention mechanism to ignore the missing modality 2.

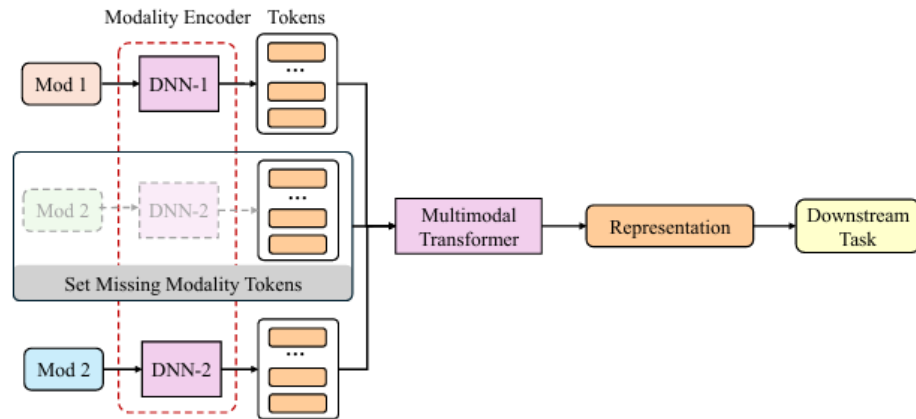


Figure 10: Description of general idea of joint representation learning methods, assuming Modality 2 is the missing modality. The missing modality 2 tokens can be replaced by specific masked tokens. The modality encoders (red dashed box) can be replaced by linear projection layers in some methods

4.1.2 Knowledge Distillation (KD) Models

Knowledge distillation transfers multi-sensory expertise from a complex "Teacher Model" (trained on complete data sets) to a streamlined "Student Model" (trained on incomplete data inputs).

Representation-Based Distillation Methods

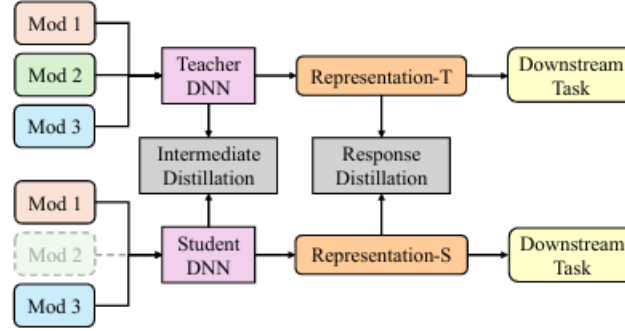


Figure 11: Description of representation distillation methods. Here we set modality 2 is missing. In order to distinguish the representations of teachers and students, we add “-T” and “-S” after their representations in the figure. We refer to intermediate distillation as using features from any combination of intermediate layers within models.

Mean Teacher Distillation (MTD): The Teacher model's parameters are calculated as an Exponential Moving Average (EMA) of the Student model's parameters over training steps, stabilizing training trajectories in highly unstable domains like emotional computing.

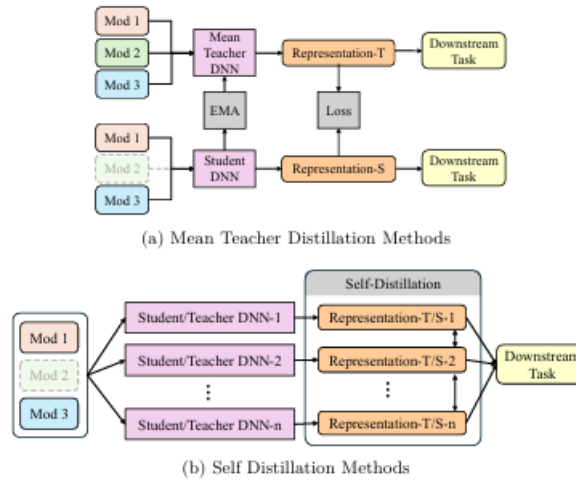


Figure 12: Description of two process-based distillation methods. (a) “EMA” means exponential moving average. To demonstrate the general architecture of Mean Teacher Distillation, we set the loss on the logits representation. (b) Self-distillation methods for missing modality problems involve the use of models that act as teachers and students, where these models use their own soft labels/representations from each branch to refine themselves during training.

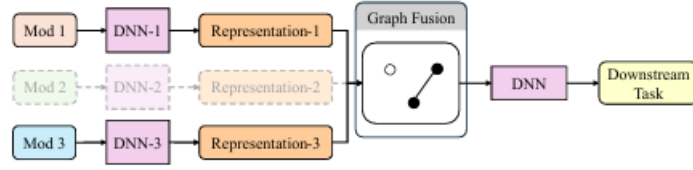


Figure 13: Illustration of graph fusion methods. We set modality 2 (white circle in the graph) as the missing modalities. Features from modality 1 and 3 can be aggregated via this graph fusion method and fused features maintains the same dimension.

Self-Distillation: The model distills knowledge internally across its own hidden branches. Complete sub-networks pass representations down to incomplete sub-paths, eliminating the need to train a separate teacher model.

4.1.3 Graph Learning-Based Fusion

Models distinct modalities as individual nodes within a topological graph, where edges represent cross-modal dependencies. Message-passing algorithms dynamically route information across remaining nodes to reconstruct or substitute the missing node's representation.

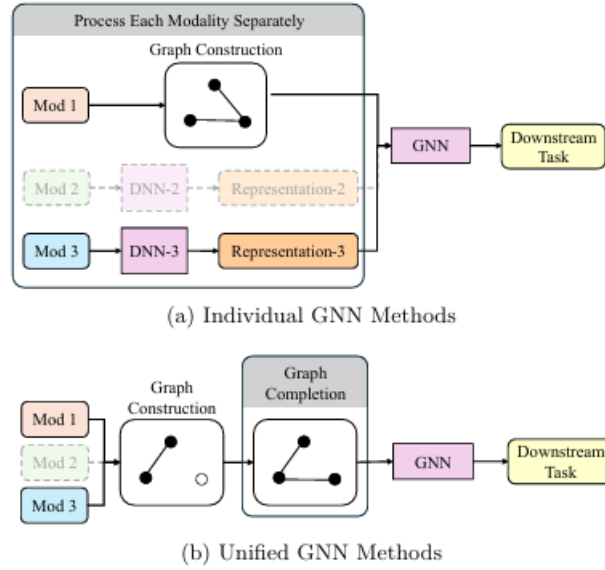


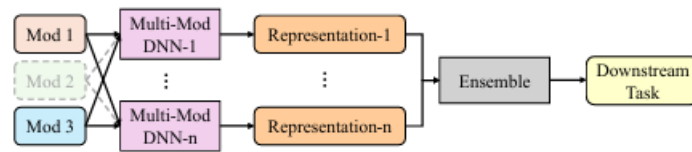
Figure 14: Illustration of graph neural network (GNN) based methods. We set modality 2 as the missing modalities. (a) Each modality first passes through DNN or builds its graph, and then they are uniformly processed using GNNs. (b) This method first complete the graph from missing-modality samples via approaches like FeatProp (Malitesta et al., 2024), then process it by GNNs.

4.1.4 Multimodal Large Language Models (MLLMs)

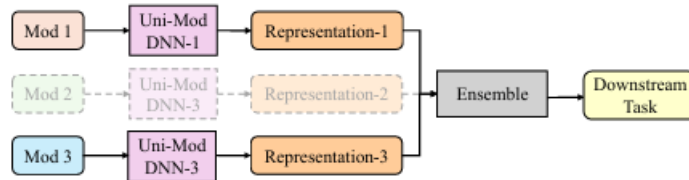
Uses foundation models as unified feature spaces. MLLMs possess vast contextual storage capacity, allowing them to parse tokenized inputs from any arbitrary subset of available sensors and maintain functional outputs.

4.2 Model Combinations

These approaches address missingness externally by organizing and executing distinct models or toolsets.



(a) Multimodal Model Ensemble Methods



(b) Unimodal Model Ensemble Methods

Figure 15: Two general architectures for ensemble methods. (a) The multimodal model ensemble methods contain n three-modal DNNs and produce final predictions based on the outputs of all n DNNs. (b) Each modality is processed by a unimodal DNN in unimodal model ensemble methods, where final predictions are produced by aggregating the outputs of all accessible unimodal DNNs.

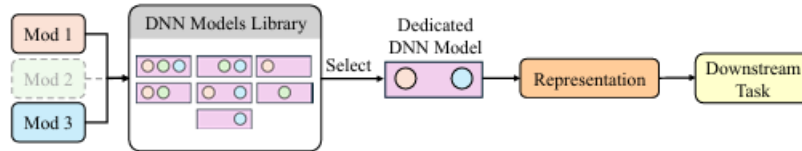


Figure 16: General idea of dedicated training methods. Assuming that modality 2 is missing, dedicated methods select the sample-suited model from DNN models library for training and testing. In the figure, the three modalities can form a total of seven models with different modality combinations. For easy understanding, we use colored circles in purple rectangles to represent the modalities that the model can handle. In order to adapt to the missing modality 2 (light green circles), dedicated methods usually select the DNN model (purple rectangle) that can handle modality 1 (light orange circles) and modality 3 (light blue circles) for training and testing

Ensemble Frameworks: Combines predictions from multiple distinct unimodal or sub-multimodal models. The final prediction is resolved via voting, averaging, or stacking networks based on which sensors are active.

Dedicated Case Training: Trains completely independent, isolated models for every single mathematical combination of available/missing modalities. While highly accurate for fixed environments, it suffers from exponential model proliferation

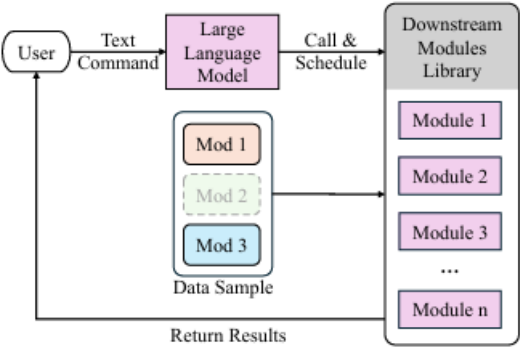


Figure 17: General procedure of discrete scheduler methods. The user types text instructions to make an LLM call and schedule the selected downstream task modules to process data samples and return results to the user. As long as there are sufficiently diverse downstream task modules, it is possible to handle the same downstream tasks with different numbers of modalities

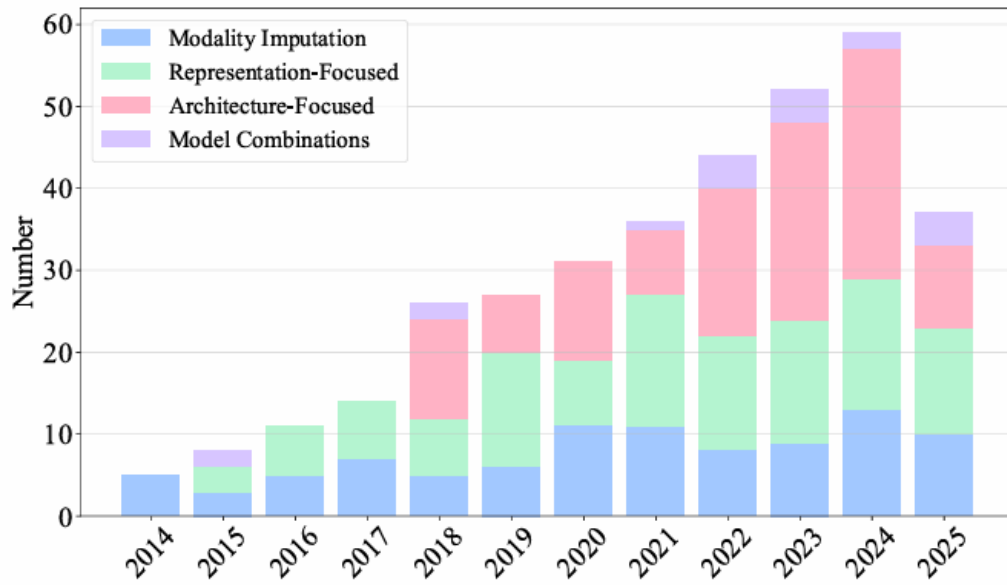
Discrete Schedulers: Employs an intelligent LLM agent as a central coordinator. The agent inspects the incoming data streams, detects missing components, and dynamically routes the execution flow to the appropriate specialized downstream tools.

5. Methodology Discussion

Aspect	Type	Method	Pros and Cons
Data Processing	Modality Imputation	Modality Composition	Pros: Simple and effective for data augmentation. Cons: Unsuitable for pixel-level tasks (Composition);
		Modality Generation	Modality number $\uparrow$, generative model number or training complexity $\uparrow$.
	Representation Focused	Coordinated-Representation	Pros: Flexible for novel modalities with better generalization.
		Representation Composition Representation Generation	Cons: Hard to balance constraints and handle dataset imbalance.
Strategy Design	Architecture Focused	Attention-based	Pros: Attention methods scale well;
		Distillation-based	Distillation and graph learning are effective.
		Graph Learning-based	Cons: Attention methods often demand large computation & datasets; Distillation and graph learning struggle with incomplete datasets and training complexity.
		Multimodal Large Language Model	
	Model Combinations	Ensemble	Pros: Effective for specific tasks.
		Dedicated	Cons: Grows complex with more modalities;
		Discrete Scheduler	Suffers from modality imbalance; Demands significant storage space.

Recovery and Non-Recovery Methods & Some Statistics

Methods	Early	Intermediate	Late	Hybrid
Recovery	Modality Composition [Yang et al. 2024] Modality Generation [Wang et al. 2024]	Representation Composition [Wang et al. 2023b] Representation Generation [Wu et al. 2023] Distillation-based [Shen & Gao 2019] Graph Learning-based [Maliresta et al. 2024]	Representation Generation [Ma et al. 2021c] Distillation-based [Li et al. 2022c] Coordinated-Representation [Ma et al. 2021b]	Distillation-based [Sikdar et al. 2023]
Non-Recovery	Dedicated [Wang et al. 2021] Discrete Scheduler [Wu et al. 2023a]	Representation Composition [Delgado-Escano et al. 2021] Attention-based [Ge et al. 2023] Graph Learning-based [Zhao et al. 2022] Multimodal Large Language Model [Wu et al. 2023b]	Coordinated-Representation [Liang et al. 2019] Distillation-based [Shi et al. 2024] Ensemble [Chen et al. 2022]	Not Applicable



6.Applications and Datasets

- **Affective Computing**
- **Medical Applications**
- **Information Retrieval**
- **Remote Sensing**
- **Pervasive Computing**
- **Robotic Vision and Sensing**

Applications	Modality Types	Common and Typical Datasets
Affective Computing	Vision+Audio	eNTERFACE'05 [Martin et al. 2006], RAVDESS [Livingstone & Russo 2018], CREMA-D [Cao et al. 2014]
	Vision+Text	eBDtheque [Guérin et al. 2013], DCAd [Nguyen et al. 2018], Manga 109 [Matsui et al. 2017]
	Vision+Bio-Sensors	Ulm-TSST [Stappen et al. 2021a], RECOLA [Ringeval et al. 2013], SEED [Zheng & Lu 2015]
	Vision+Audio+Text	CMU-MOSI [Zadeh et al. 2016], CMU-MOSEI [Zadeh et al. 2018], IEMOCAP [Busso et al. 2008], MSP-IMPROV [Busso et al. 2016], MELD [Poria et al. 2018], CH-SIMI [Yu et al. 2020], ICT-MMMO [Wollmer et al. 2013], YouTube [Morency et al. 2011]
Medical Applications	MRI, CT Scans	BRATS-series [Menze et al. 2014], Myronenko 2019, ADNI [Jack Jr et al. 2008], IXI [https://brain-development.org/ixi-dataset], StressID [Chapelonkay et al. 2024], PhysioNet [Kenop et al. 2000], TCGA [https://portal.gdc.cancer.gov/], BCData [Huang et al. 2020], NsFlick [Koohehnamini et al. 2020], LYON19 [Swiderska-Chadaj et al. 2019]
	Bio-Sensors	MIMIC-CXR [Johnson et al. 2019], MIMIC-IV [Johnson et al. 2023]
	Electronic Health Record	NCH [Lee et al. 2022], BC5B [Xu et al. 2021], ODIR [Li et al. 2021]
Information Retrieval	Vision+Text	Amazon Series Datasets [Lakkaraju et al. 2013], McAuley et al. [2015], He & McAuley [2016], PromptCloud [2017]
	Vision+Audio+Text	MIR-Flickr25K [Duin], NUS-WIDE [Cana et al. July 8-10, 2009], MSR-VTT [Xu et al. 2016], TikTok [Lin et al. 2023]
Remote Sensing	Vision	WHU-OPT-SAR [Li et al. 2022b], DFC2020 [Yokova et al. 2020], MSAW [Shermeyer et al. 2020], Trento [Rasti et al. 2017], Houston [Piacentini 2023a], Berlin [Hong et al. 2021], GRSS [Debes et al. 2014]
Pervasive Computing	Motion sensors	Opportunity [Roggen et al. 2010], RealDisp [Banos et al. 2014], DSADS [Barshan & Altun 2010], PAMAP2 [Reiss 2012], Shoaib [Shoaib et al. 2013], RealWorld-HAR [Sztaylor & Stuckenschmidt 2016], w-HAR [Burzer et al. 2025], Epilepsy [Villar et al. 2016], EMG Physical Action [Theodoridis 2011], ERing [Wilhelm et al. 2015], and GENEActiv [Phillips et al. 2013]
	Physiological Signals	WESAD [Schmidt et al. 2018a], mibran21 [Donckx 2024], ETRI Litesog [Oh et al. 2023]
	Electroencephalogram	Self RegulationSPT [Rebaumer et al. 1999], ECG-MIXD [Scholk et al. 2001], RegDB [Ye et al. 2021], SYSU-MM01 [Wu et al. 2017], UWAIDII [Kahmann et al. 2016]
Robotic Vision and Sensing	Vision	Delivery [Zhang et al. 2023a], NuScenes [Cassara et al. 2020], MVSS [Ji et al. 2023]
	Vision+Audio	Ego4D-AR [Grauman et al. 2022], Epic-Sounds [Huh et al. 2023], Epic-Kitchens [Janen et al. 2022], Speaking Faces [John & Kawanishi 2022]
	Vision+Skeleton/Inertial Data	MMG-Ego4D [Gong et al. 2023], Northwestern-UCLA [Wang et al. 2014], NTU60 [Shahroudy et al. 2016]

6. Open Challenges & Future Horizons

The Recovery vs. Non-Recovery Dilemma: A key problem is determining whether to synthesize the missing channel or compute using only the remaining data. Generative approaches risk creating false artifacts (hallucinations), while non-recovery approaches can allow dominant modalities to overpower subtle but critical inputs.

Lack of Evaluation Standardization: Current research lacks uniform evaluation benchmarks. Studies apply varying missing ratios (e.g., 20% to 80%) under different patterns (random vs. structured), making direct architectural comparisons difficult.

Real-Time Asynchronous Streaming: Real-world deployments (e.g., autonomous driving, medical monitoring) require handling dynamic, asynchronous, and transient data dropouts rather than static missingness, presenting a vital path for future model development.